

# Dequantization of Real Algebraic Geometry on a Logarithmic Paper

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**Abstract.** On logarithmic paper some real algebraic curves look like smoothed broken lines. Moreover, the broken lines can be obtained as limits of those curves. The corresponding deformation can be viewed as a quantization, in which the broken line is a classical object and the curves are quantum. This generalizes to a new connection between algebraic geometry and geometry of polyhedra, which is more straight-forward than the other known connections and gives a new insight into constructions used in topology of real algebraic varieties.

## 1. Graphs of Polynomials on a Logarithmic Paper

### 1.1. How to Visualize a Real Polynomial?

If you ever tried to draw the graph for a polynomial of degree greater than, say, 4 and consisting of at least 4 monomials, you are aware about the natural difficulties. The graph is too steep. Whatever scale you choose, either some important details do not fit to the picture or are too small. The usual recipes from Calculus do not address the problem, but suggest, instead, to find roots of the first two derivatives, which does not seem to be much easier than the original problem.

### 1.2. Logarithmic Paper

A physicist or engineer can give a more practical advise, based on their experience: *use a (double) logarithmic paper*. This is a graph paper, called also a *log paper*, with a non-uniform net of coordinate lines and logarithmic scales on both axes. On a log paper a point with coordinates  $x, y$  is shown at the position with the usual, Cartesian coordinates equal to  $\ln x, \ln y$ . In other words, the transition to the log paper corresponds to the change of coordinates:

$$\begin{cases} u = \ln x \\ v = \ln y. \end{cases}$$

On a log paper the first quadrant is expanded homeomorphically to the whole plane, the line  $x = 1$  occupies the position of the axis of ordinates, the line  $y = 1$  occupies the position of the axis of abscissas, the unit square bounded by these lines and the coordinate axes occupies the whole third quadrant.

### 1.3. A Monomial on a Logarithmic Paper

Let us try to follow the advise to use a log paper. Consider first the simplest special case: the graph of a monomial  $ax^k$  (i. e., the curve defined by  $y = ax^k$ ). We are forced to consider only positive  $x, y$  and hence assume  $a$  to be positive as well. Then  $v = \ln y = \ln(ax^k) = k \ln x + \ln a = ku + \ln a = ku + b$ , where we denote  $\ln a$  by  $b$ . Now the curve which we want to draw is defined in coordinates  $u$  and  $v$  by equation  $v = ku + b$ . Everybody knows that this is the straight line with slope  $k$  meeting the axis of ordinates (i.e.,  $v$ -axis) at  $(0, b)$ .

### 1.4. Few Slightly More Complicated Polynomials

First, consider a line  $y = 1 + x$ . Then

$$v = \ln y = \ln(1 + x) = \ln(1 + e^u).$$

See the left plot in Figure 1, where the graph of  $v = \ln(1 + e^u)$  is shown together with lines  $v = 0$  and  $v = u$ , which represent on the log paper the monomials 1 and  $x$  involved in our polynomial  $1 + x$ . The graph of  $v = \ln(1 + e^u)$  looks like the broken line  $v = \max\{0, u\}$  with a smoothed corner: it goes along and above of this broken line getting very close to it as  $|u|$  grows. For  $|u| > 4$  the difference becomes beyond the resolution.

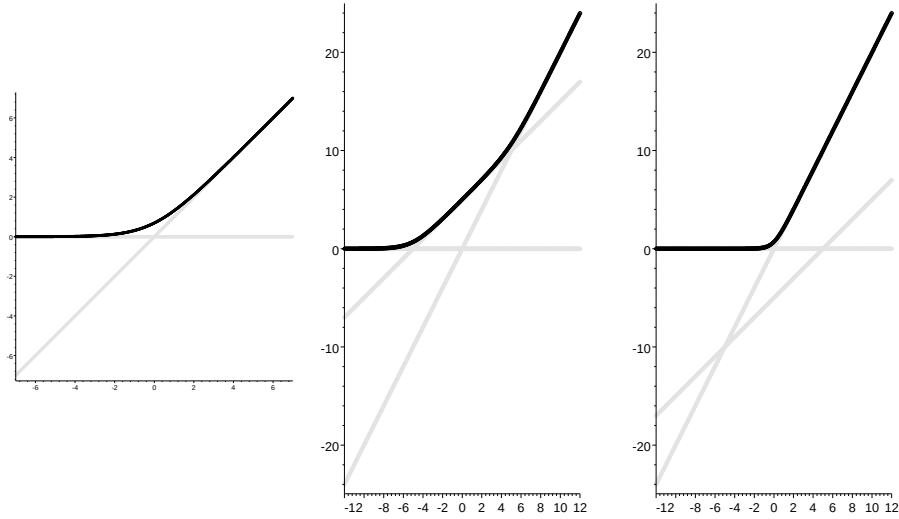


FIGURE 1. Log paper graphs of  $1 + x$ ,  $1 + e^5x + x^2$  and  $1 + e^{-5}x + x^2$ .

Here are further two examples: quadratic polynomials  $1 + e^{\pm 5}x + x^2$ . Then

$$v = \ln(1 + e^{\pm 5}x + x^2) = \ln(1 + e^{u \pm 5} + e^{2u}).$$

See the central and right plots in Figure 1. The graph of  $v = \ln(1 + e^{u \pm 5} + e^{2u})$  looks like the broken line  $v = \max\{0, u \pm 5, 2u\}$  with smoothed corners. It goes along and above of this broken line getting very close to it far from its corners. Notice that the lines  $v = 0$ ,  $v = u \pm 5$  and  $v = 2u$  represent on the logarithmic paper the monomials  $1$ ,  $e^{\pm 5}x$  and  $x^2$ , respectively.

### 1.5. A Polynomial Versus the Maximum of Its Monomials

This suggests, for a polynomial  $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$  with positive real coefficients  $a_i = e^{b_i}$ , to compare the graphs on a log paper for  $p$  and the maximum  $M(p)(x) = \max\{a_n x^n, a_{n-1} x^{n-1}, \dots, a_0\}$  of its monomials. Denote the graph on a log paper of a function  $f$  by  $\Gamma_f$ . With respect to the usual Cartesian coordinates,  $\Gamma_p$  is the graph of

$$L_p(u) = \ln \left( e^{nu+b_n} + e^{(n-1)u+b_{n-1}} + \dots + e^{b_0} \right)$$

and  $\Gamma_{M(p)}$  is the graph of a piecewise linear convex function

$$M_p(u) = \max \{ nu + b_n, (n-1)u + b_{n-1}, \dots, b_0 \}.$$

Obviously,  $M_p(u) \leq L_p(u) \leq M_p(u) + \ln(n+1)$ . Hence  $\Gamma_p$  is above  $\Gamma_{M(p)}$ , but below a copy of  $\Gamma_{M(p)}$  shifted upwards by  $\ln(n+1)$ . The latter is in fact a rough estimate. It turns to equality only at  $u$ , where all linear functions, whose maximum is  $M_p(u)$ , are equal:  $nu + b_n = (n-1)u + b_{n-1} = \dots = b_0$ .

For a generic value of  $u$ , only one of these functions is equal to  $M_p(u)$ . Say  $M_p(u) = ku + b_k$ , while  $M_p(u) > d + lu + b_l$  for some positive  $d$  and each  $l \neq k$ . Then

$$L_p(u) < M_p(u) + \ln(1 + ne^{-d}) < M_p(u) + e^{-d}n.$$

If for some value of  $u$  the values of all of the functions  $ku + b_k$  except two are smaller than  $M_p(u) - d$ , then

$$L_p(u) < M_p(u) + \ln(2 + (n-1)e^{-d}) < M_p(u) + \ln 2 + e^{-d}(n-1)/2.$$

Thus, on a logarithmic paper the graph of a generic polynomial with positive coefficients lies in a narrow strip along the the broken line which is the graph of the maximum of its monomials. The width of the strip is estimated by characteristics of the mutual position of the lines which are the graphs of the monomials. The less congested the configuration of these lines, the narrower this strip.

### 1.6. Rescalings Pushing the Graph of a Polynomial to a PL-Graph

A natural way to make a configuration of lines less congested without changing of its topology is to apply a dilation  $(u, v) \mapsto (Cu, Cv)$  with a large  $C > 0$ . In what follows it is more convenient to use instead of  $C$  a parameter  $h$  related to  $C$  by  $h = 1/C$ . In terms of  $h$  the dilation acts by  $(u, v) \mapsto (u/h, v/h)$ . It maps the graph of  $v = ku + b$  to the graph of  $v = ku + b/h$ . The parallel operation on monomials replaces  $ax^k$  by  $a^{1/h}x^k$ .

Consider the corresponding family of polynomials:  $p_h(x) = \sum_k a_k^{1/h} x^k$ . On the log paper, the graphs of its monomials are obtained by dilation with ratio  $1/h$  from the graphs of the corresponding monomials of  $p$ . Hence  $\Gamma_{M(p_h)}$  is the image of  $\Gamma_{M(p)}$  under the same dilation. However,  $\Gamma_{p_h}$  is not the image of  $\Gamma_p$ . It still lies in a strip along  $\Gamma_{M(p_h)}$  and the strip is getting narrower as  $h$  decreases, but at the corners of  $\Gamma_{M(p_h)}$  the width of the strip cannot become smaller than  $\ln 2$ .

To keep the picture of our expanding configuration of lines (the graphs of monomials) independent on  $h$ , let us make an additional calibration of coordinates: set  $u_h = hu = h \ln x$ ,  $v_h = hv = h \ln y$ . Denote by  $\Gamma_f^h$  the graph of a function  $y = f(x)$  in the plane with coordinates  $u_h, v_h$ .

Then  $\Gamma_{M(p_h)}^h$  does not depend on  $h$ . The additional scaling reduces the width of the strip along  $\Gamma_{M(p_h)}^h$ , where  $\Gamma_{p_h}^h$  lies, forcing the width to tend to 0 as  $h \rightarrow 0$ . Thus  $\Gamma_{p_h}^h$  tends to  $\Gamma_{M(p_h)}^h$  (in the  $C^0$  sense) as  $h \rightarrow 0$ .

## 2. Quantization

### 2.1. Maslov Dequantization of Positive Real Numbers

The rescaling formulas  $u_h = h \ln x$ ,  $v_h = h \ln y$  bring to mind formulas related to the Maslov dequantization of real numbers, see e.g. [4], [5]. The core of the Maslov dequantization is a family of semirings  $\{S_h\}_{h \in [0, \infty)}$  (recall that a semiring is a sort of ring, but without subtraction). As a set, each of  $S_h$  is  $\mathbb{R}$ . The semiring operations  $\oplus_h$  and  $\odot_h$  in  $S_h$  are defined as follows:

$$(1) \quad a \oplus_h b = \begin{cases} h \ln(e^{a/h} + e^{b/h}), & \text{if } h > 0 \\ \max\{a, b\}, & \text{if } h = 0 \end{cases}$$

$$(2) \quad a \odot_h b = a + b$$

These operations depend continuously on  $h$ . For each  $h > 0$  the map

$$D_h : \mathbb{R}_+ \setminus \{0\} \rightarrow S_h : x \mapsto h \ln x$$

is a semiring isomorphism of  $\{\mathbb{R}_+ \setminus \{0\}, +, \cdot\}$  onto  $\{S_h, \oplus_h, \odot_h\}$ , that is

$$D_h(a + b) = D_h(a) \oplus_h D_h(b), \quad D_h(ab) = D_h(a) \odot_h D_h(b).$$

Thus  $S_h$  with  $h > 0$  can be considered as a copy of  $\mathbb{R}_+ \setminus \{0\}$  with the usual operations of addition and multiplication. On the other hand,  $S_0$  is a copy of  $\mathbb{R}$  where the operation of taking maximum is considered as an addition, and the usual addition, as a multiplication.

Applying the terminology of quantization to this deformation, we must call  $S_0$  a classical object, and  $S_h$  with  $h \neq 0$ , quantum ones. The analogy with Quantum Mechanics motivated the following *correspondence principle* formulated by Litvinov and Maslov [4] as follows:

*“There exists a (heuristic) correspondence, in the spirit of the correspondence principle in Quantum Mechanics, between important, useful and interesting constructions and results over the field of real (or complex) numbers (or the semiring*

of all nonnegative numbers) and similar constructions and results over idempotent semirings.”

This principle proved to be very fruitful in a number of situations, see [4], [5]. According to the correspondence principle, the idempotent counterpart of a polynomial  $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$  is a convex PL-function  $M_p(u) = \max \{nu + b_n, (n-1)u + b_{n-1}, \dots, b_0\}$ . As we have seen above,  $p$  and  $M_p$  are related not only on an heuristic level. In Section 1.6 we connected the graph  $\Gamma_p$  of  $p$  on a logarithmic paper and the graph  $\Gamma_{M(p)}$  of  $M_p$  by a continuous family of graphs  $\{\Gamma_{p_h}^h\}_{h \in (0,1)}$ .

## 2.2. Logarithmic Paper as a Graphical Device for the Maslov Dequantization

Drawing a graph on a log paper can be thought of as applying of  $D_1 \times D_1 : (\mathbb{R}_+ \setminus 0)^2 \rightarrow S_h^2 = \mathbb{R}^2$  to the Cartesian graph. The log paper scaled with ratio  $1/h$ , which appeared in Section 1.6, is nothing but  $S_h^2$ , and  $\Gamma_f^h$  is the image of the graph of  $f$  under the map  $D_h \times D_h : (\mathbb{R}_+ \setminus 0)^2 \rightarrow S_h^2 = \mathbb{R}^2$ . Furthermore,  $\Gamma_{p_h}^h$  is the graph of the function  $S_h \rightarrow S_h$  defined by polynomial  $p_h(x) = \sum_k a_k^{1/h} x^k$  evaluating via the operations  $\oplus_h$  and  $\odot_h$  of semiring  $S_h$ .

In terms of the native coordinates  $u, v$  in  $S_h^2 = \mathbb{R}^2$  (which were denoted in Section 1.6 by  $u_h, v_h$ ) and the classical functions of them,  $\Gamma_{p_h}^h$  is defined by equation  $v = h \ln(\sum_k e^{(ku+b_k)/h})$ . The right hand side of this equation tends to  $\max \{ku + b_k \mid k = 0, \dots, n\} = M_p(u)$  as  $h \rightarrow 0$ . In terms of the semiring operations in  $S_0$ , this limit is the value of polynomial  $\sum_k b_k x^k$ .

## 2.3. Dequantization of a Polynomial

For any real polynomial  $p(x) = \sum_k a_k x^k$  with positive coefficients we have defined above (in Section 1.6)  $p_h(x) = \sum_k a_k^{1/h} x^k$  for  $h > 0$ . Extend this definition to  $h = 0$  by setting  $p_0(x) = \sum_k \ln(a_k) x^k$ .

Denote by  $p(u)_h$  the result of evaluation of a polynomial  $p$  with positive real coefficients at  $u \in S_h = \mathbb{R}$  by the semiring operations of  $S_h$  defined by (1) and (2). Considerations of Section 1.6 reformulated at the end of Section 2.2 immediately imply continuity of the function  $\mathbb{R}_+ \times \mathbb{R} = \cup_{h \in \mathbb{R}_+} S_h \rightarrow S_h = \mathbb{R} : (h, u) \mapsto p_h(u)_h$ .

Thus we can consider  $p_h$  as a natural continuous family of polynomials connecting  $p = p_1$  to its idempotent counterpart  $p_0$ .

## 2.4. Real Algebraic Geometry as Quantized PL-Geometry

The notion of polynomial is central in the algebraic geometry. (I believe, the subject of algebraic geometry would be better described by the name of *polynomial geometry*.) Since a polynomial over  $\mathbb{R}$  is presented so explicitly as a quantization of a piecewise linear convex function, one may expect to find along this line explicit relations between other objects and phenomena of algebraic geometry over  $\mathbb{R}$  and piecewise linear geometry. Indeed, in the piecewise linear geometry the notion of piecewise linear convex function plays almost the same rôle as the notion of polynomial in the algebraic geometry.

A representation of real algebraic geometry as a quantized PL-geometry may be rewarding in many ways. For example, in any quantization there are *classical objects*, i.e. objects which do not change much under the quantization. Objects of PL-geometry are easier to construct. If we knew conditions under which a PL object gives rise to a real algebraic object, which is classical with respect to the Maslov quantization, then we would have a simple way to construct real algebraic objects with controlled properties.

### 3. Put Algebraic Geometry Onto a Logarithmic Paper

#### 3.1. Speak on Real Algebraic Geometry Only Positively

Above in Section 1 we discussed drawing of graph on a logarithmic paper only for a polynomial with positive coefficients. The graphs allowed us to see the behaviour of the polynomials only at positive values of the argument. This was for a good reason: we used logarithms of coordinates. The Maslov dequantization deals only with positive numbers. Therefore each fragment of algebraic geometry, that we want to dequantize, must be reformulated first only in terms of positive numbers. This seems to be possible for everything belonging to the real algebraic geometry.

#### 3.2. Visualising Roots of a Polynomial on a Logarithmic Paper

Above we could not meet roots of polynomials, for a polynomial with *positive* coefficients has no *positive* roots. However if we really want to do the algebraic geometry on the log paper, we must figure out how to use graphs on the log paper for visualising roots (well, only *positive* roots) of an arbitrary real polynomial.

Any real polynomial  $p(x)$  is a difference  $p^+(x) - p^-(x)$  of polynomials with positive coefficients. We can reformulate the problem of finding the positive roots of  $p$  as the problem of finding positive values of  $x$  at which  $p^+(x) = p^-(x)$ . The graphs of  $p^+$  and  $p^-$  can be drawn on a log paper, where they are localised in the strips along broken lines, see Section 1.5 above. For some polynomials this picture gives a decent information on the number and position of the positive roots.

The negative roots of  $p(x)$  can be treated in the same way, since their absolute values are the positive roots of  $p(-x)$ .

#### 3.3. Plane Algebraic Curves on a Logarithmic Paper

Now consider a real polynomial  $p(x, y) = \sum_{k,l} a_{k,l} x^k y^l$  in two variables. Similarly to the case of polynomials in one variable, in the *logarithmic space*  $S_1^3$  the graph of a monomial  $ax^k y^l$  with  $a > 0$  is a plane  $w = ku + lv + \ln a$ , and the graph  $\Gamma_p$  of a polynomial  $p(x, y)$  with positive coefficients lies in a neighborhood of a convex piecewise linear surface, which is the graph  $\Gamma_{M(p)}$  of the maximum  $M(p)$  of the monomials. Furthermore,  $p$  is included into a dequantizing family  $p_h$  defined as  $\sum_{k,l} a_{k,l}^{1/h} x^k y^l$  for  $h > 0$  and  $\sum_{k,l} \ln(a_{k,l}) x^k y^l$  for  $h = 0$ , cf. Section 2.3. The graphs  $\Gamma_{p_h}^h$  of  $p_h$  in  $S_h^3$  comprise a continuous deformation of  $\Gamma_p = \Gamma_{p_1}^1$  to  $\Gamma_{M(p)} = \Gamma_{p_0}^0$ .

For a polynomial  $p$  in two variables with arbitrary real coefficients, denote by  $p^+$  the sum of its monomials with positive coefficients, and put  $p^- = p^+ - p$ . Thus  $p$  is canonically presented as a difference  $p^+ - p^-$  of two polynomials with positive coefficients. To obtain the curve defined on a logarithmic paper by the equation  $p(x, y) = 0$ , one can construct the graphs  $\Gamma_{p^+}$  and  $\Gamma_{p^-}$  for  $p^+$  and  $p^-$  in the logarithmic space, which are the surfaces defined in the usual Cartesian coordinates by  $w = \ln(p^\pm(e^u, e^v))$ , and project the intersection  $\Gamma_{p^+} \cap \Gamma_{p^-}$  to the plane of arguments.

For the first approximation of this curve, one may take the broken line, which is the projection of the intersection of the piecewise linear surfaces  $\Gamma_{M(p^+)}$  and  $\Gamma_{M(p^-)}$  corresponding to  $p^+$  and  $p^-$ .

Of course, it may well happen that the broken line does not even resemble the curve. This happens to first approximations. However, it is very appealing to figure out circumstances, under which the broken line is a good approximation, for a broken line seems to be much easier to deal with, than an algebraic curve.

### 3.4. Constructing Algebraic Curves Classical From Our Quantum Point of View

Recall that in the logarithmic space  $S_h^3$  the graph of  $ax^k y^l$  is a plane  $w = ku + lv + \ln a$ . It has a normal vector  $(k, l, -1)$  and intersects the vertical axis at  $(0, 0, \ln a)$ . Thus if we want to construct a curve of a given degree  $m$ , we have to arrange planes whose normals are fixed: they are  $(k, l, -1)$  with integers  $k, l$ , satisfying inequalities  $0 \leq k, l, k + l \leq m$ . The only freedom is in moving them up and down.

Consider the pieces of these planes which do not lie under the others. They form a convex piecewise linear surface  $U$ , the graph of the maximum of the linear forms defining our planes. We will call  $U$  the *upper surface* of the configuration. The combinatorial structure of faces in  $U$  depends on the arrangement. Assume that at each vertex of  $U$  exactly three of the planes meet. This is a genericity condition, which can be satisfied by small shifts of the planes.

Divide now the faces of  $U$  arbitrarily into two classes. Denote the union of one of them by  $U^+$ , the union of the other by  $U^-$ . By genericity of the configuration, the common boundary of  $U^+$  and  $U^-$  is a union of disjoint polygonal simple closed curves. It can be easily realized as  $C_{M(p)}$ : take for  $p^\varepsilon$  with  $\varepsilon = \pm$  the sum of monomials corresponding to the planes of faces forming  $U^\varepsilon$  and put  $p = p^+ - p^-$ .

Consider now the family of curves  $C_{p_h} = \Gamma_{p_h^+}^h \cap \Gamma_{p_h^-}^h \subset S_h^3$  with  $1 \geq h \geq 0$ . At  $h = 0$  this is the intersection of the convex PL-surfaces  $\Gamma_{M(p^+)}$ ,  $\Gamma_{M(p^-)}$ . Due to the genericity condition above, this intersection is as transversal as one could wish: at all but a finite number of points an interior part of a face of one of them meets an interior part of a face of the other one, and at the rest of points an edge of one of the surfaces intersects transversally an interior of a face of the other surface.

At a point of the first kind, transversality is obviously preserved, when  $h$  gets positive, since there each of  $p_h^+$  and  $p_h^-$  has a dominating monomial and the gradients of these monomials are not parallel. At a point of the second kind, one of the polynomials has two dominating monomials (corresponding to the two faces

adjacent to the edge), their gradients vanish in the direction of the common edge and this edge is transversal to the other surface, since it intersects transversally the face of the PL-surface corresponding to the dominating monomial.

Take the curve corresponding to a value of  $h$  such that the transversality is preserved between 0 and this value. The projection to  $(u, v)$ -plane of  $C_{p_h}$  represents an algebraic curve of degree  $m$  on the scaled logarithmic paper  $S_h^2$  and it can be obtained by a small isotopy of the projection of  $\partial U^+$  to the  $(u, v)$ -plane.

### 3.5. Is This a Patchworking?

A construction, which looks similar, has been known in the topology of real algebraic varieties for about 20 years as a *patchworking*, or *Viro's method*. It has been used to construct real algebraic varieties with controlled topology and helped to solve a number of problems. For example, to classify up to isotopy non-singular real plane projective curves of degree 7 [7], [8] and disprove the Ragsdale conjecture [2] on topology of plane curves formulated [6] as early as in 1906. To the best of my knowledge, the patchworking has never been related to the Maslov quantization.

## 4. Patchworking Real Algebraic Curves

### 4.1. The Simplest Patchworking

Here is a description of a simplified version of the patchworking. The simplifications are of the following 3 kinds:

- we restrict to the case of nonsingular planar curves,
- we assume that all patches are trinomials, and
- we consider only the part of the curve contained in the first quadrant (what happens in other quadrants is described soon after).

**Initial Data.** Let  $m$  be a positive integer (it will be the degree of the curve under construction) and  $\Delta$  be the triangle in  $\mathbb{R}^2$  with vertices  $(0, 0)$ ,  $(m, 0)$ ,  $(0, m)$ . Let  $\tau$  be a *convex* triangulation of  $\Delta$  with vertices having integer coordinates. Convexity of  $\tau$  means that there exists a convex piecewise linear function  $\nu : \Delta \rightarrow \mathbb{R}_+$  which is linear on each triangle of  $\tau$  and is not linear on the union of any two triangles of  $\tau$ . Let the vertices of  $\tau$  be equipped with signs. The sign (plus or minus) at the vertex with coordinates  $(k, l)$  is denoted by  $\sigma_{k,l}$ .

**Construction of Piecewise Linear Curve.** If a triangle of the triangulation  $\tau$  has vertices of different signs, draw a midline separating pluses from minuses. Denote by  $L$  the union of these midlines. It is a collection of polygonal lines contained in  $\Delta$ . The pair  $(\Delta, L)$  is called the *result of combinatorial patchworking*.

**Construction of Polynomials.** Given initial data  $m$ ,  $\Delta$ ,  $\tau$  and  $\sigma_{k,l}$  as above and a positive convex function  $\nu$  certifying, as above, that the triangulation  $\tau$  is



convex. Consider a one-parameter family of polynomials

$$(3) \quad b_t(x, y) = \sum_{\substack{(k, l) \text{ runs over} \\ \text{vertices of } \tau}} \sigma_{k, l} t^{\nu(k, l)} x^k y^l.$$

Polynomials  $b_t$  are called the results of *polynomial patchworking*.

**Patchwork Theorem.** *Let  $m, \Delta, \tau, \sigma_{k, l}$  and  $\nu$  be initial data as above. Denote by  $b_t$  polynomials obtained by the polynomial patchworking of these initial data, and by  $L$  the PL-curve in  $\Delta$  obtained from the same initial data by the combinatorial patchworking.*

*Then for all sufficiently small  $t > 0$  the polynomial  $b_t$  defines in the first quadrant  $\mathbb{R}_{++}^2 = \{(x, y) \in \mathbb{R}^2 \mid x, y > 0\}$  a curve  $a_t$  such that the pair  $(\mathbb{R}_{++}^2, a_t)$  is homeomorphic to the pair  $(\text{Int } \Delta, L \cap \text{Int } \Delta)$ .*

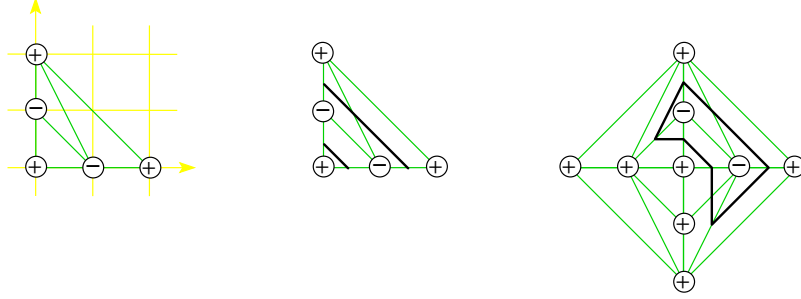


FIGURE 2. Patchworking: initial data, construction of the PL-curve in the first quadrant and on the whole plane. The corresponding algebraic curves are ellipses meeting the coordinate axes in their positive halves.

#### 4.2. Patchwork in Other Quadrants

Patchwork Theorem applied to  $b_t(-x, y)$ ,  $b_t(x, -y)$  and  $b_t(-x, -y)$  gives a similar topological description of the curve defined in the other quadrants by  $b_t$  with sufficiently small  $t > 0$ . The results can be collected in the following natural combinatorial construction.

**Construction of PL-Curve.** Take copies  $\Delta_x = s_x(\Delta)$ ,  $\Delta_y = s_y(\Delta)$ ,  $\Delta_{xy} = s(\Delta)$  of  $\Delta$ , where  $s_x, s_y$  are reflections with respect to the coordinate axes and  $s = s_x \circ s_y$ . Denote by  $A\Delta$  the square  $\Delta \cup \Delta_x \cup \Delta_y \cup \Delta_{xy}$ . Extend the triangulation  $\tau$  to a symmetric triangulation of  $A\Delta$ , and the distribution of signs  $\sigma_{i, j}$  to a distribution at the vertices of the extended triangulation by the following rule:  $\sigma_{i, j} \sigma_{\varepsilon i, \delta j} \varepsilon^i \delta^j = 1$ , where  $\varepsilon, \delta = \pm 1$ . In other words, passing from a vertex to its mirror image with respect to an axis we preserve its sign if the distance from the vertex to the axis is even, and change the sign if the distance is odd.

If a triangle of the triangulation of  $A\Delta$  has vertices of different signs, select (as above) a midline separating pluses from minuses. Denote by  $AL$  the union of the selected midlines. It is a collection of polygonal lines contained in  $A\Delta$ . The pair  $(A\Delta, AL)$  is called the *result of affine combinatorial patchworking*. Glue by  $s$  the sides of  $A\Delta$ . The resulting space  $P\Delta$  is homeomorphic to the real projective plane  $\mathbb{R}P^2$ . Denote by  $PL$  the image of  $AL$  in  $P\Delta$  and call the pair  $(P\Delta, PL)$  the *result of projective combinatorial patchworking*.

**Addendum to Patchwork Theorem.** *Under the assumptions of Patchwork Theorem above, for all sufficiently small  $t > 0$  there exist a homeomorphism  $A\Delta \rightarrow \mathbb{R}^2$  mapping  $AL$  onto the affine curve defined by  $b_t$  and a homeomorphism  $P\Delta \rightarrow \mathbb{R}P^2$  mapping  $PL$  onto the projective closure of this affine curve.*

#### 4.3. The Simplest Patchworking Coincides With Construction of Section 3.4

The polynomial  $b_t$  defined by (3) is presented as  $b_t^+ - b_t^-$ , where

$$b_t^\varepsilon(x, y) = \sum_{\substack{(k, l) \text{ runs over the vertices} \\ \text{of } \tau, \text{ at which } \sigma_{k, l} = \varepsilon}} t^{\nu(k, l)} x^k y^l$$

Observe that polynomials  $b_t^\pm$  comprise dequantizing families. Indeed, if we take  $p(x, y) = \sum_{k, l} a_{k, l} x^k y^l$  with  $a_{k, l} = e^{-\nu(k, l)}$ , then for  $h = -1/\ln t$  we obtain

$$p_h(x, y) = \sum_{k, l} a_{k, l}^{1/h} x^k y^l = \sum_{k, l} e^{-\nu(k, l)/h} x^k y^l = \sum_{k, l} e^{\nu(k, l) \ln t} x^k y^l = \sum_{k, l} t^{\nu(k, l)} x^k y^l.$$

Patchwork Theorem deals with sufficiently small positive  $t$ , while  $h$  in a dequantizing family of polynomials was a small positive number approaching 0. This is consistent with our setting  $h = -1/\ln t$ .

A monomial  $a_{k, l} x^k y^l = e^{-\nu(k, l)} x^k y^l$  is presented in the logarithmic space by the graph of  $w = ku + lv - \nu(k, l)$ . Hence the graph of the maximum of linear forms corresponding to all monomials of  $p^+$  and  $p^-$  is defined by

$$(4) \quad w = \max\{ku + lv - \nu(k, l) \mid (k, l) \text{ runs over vertices of } \tau\}.$$

In (4) we recognise the convex function conjugate to  $\nu$ . The graph of (4) is a convex PL surface, whose natural stratification is dual to the triangulation  $\tau$  of  $\Delta$ : the face which lies on the plane  $w = ku + lv - \nu(k, l)$  corresponds to the vertex  $(k, l)$  of  $\tau$ , two such faces meet at an edge in the graph of (4) iff the corresponding vertices are connected with an edge of  $\tau$ , three faces meet at a vertex iff the corresponding vertices of  $\tau$  belong to a triangle of  $\tau$ . In particular, we see that the configuration of planes satisfies the genericity condition of Section 3.4 and planes  $w = ku + lv - \nu(k, l)$  corresponding to all monomials of  $b_t^\pm$  show up in the graph of (4) as its faces.

Some of these faces correspond to monomials of  $b_t^+$ , the others to monomials of  $b_t^-$ . The edges which separate the faces of these two kinds comprise a broken line  $C_{M(p)}$ , cf. Section 3.4. These edges are dual to the edges of  $\tau$  which intersect the result  $L$  of the combinatorial patchworking.

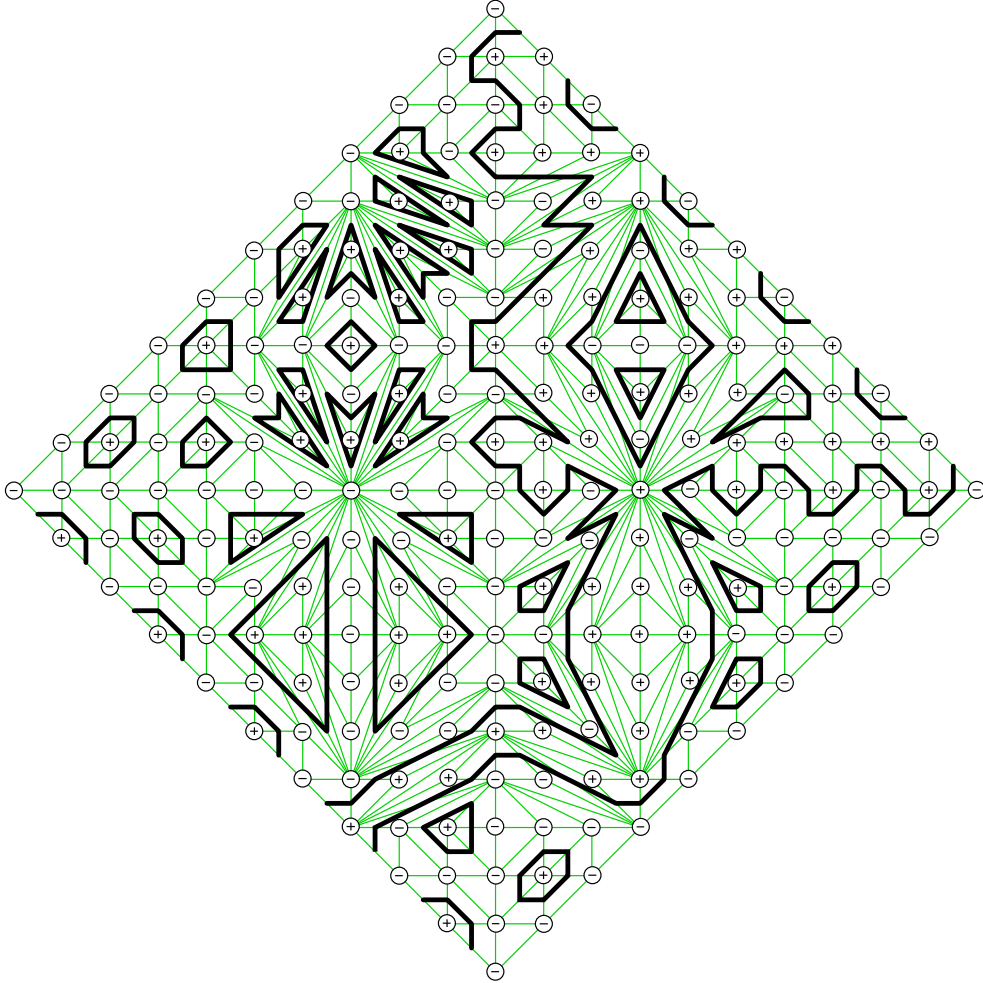


FIGURE 3. Patchworking of a counter-example to the Ragsdale Conjecture. A curve of degree 10 with 32 odd ovals constructed by Itenberg [2].

Therefore the topology of the projection to the  $(u, v)$ -plane of  $C_{M(p)}$  coincides with the topology of  $L$  in  $\Delta$ .  $\square$

### Conclusion

We see that the quantum point of view (or its graphical log paper equivalent) gives a natural explanation to the simplest patchwork construction. The proofs become more conceptual and straight-forward. Of course, similar but slightly more involved quantum explanations can be given to all versions of patchwork.

Let me shortly mention other problems which can be attacked using similar arguments.

First of all, this is a Fewnomial Problem. Although A. G. Khovansky [3] proved that basically all topological characteristics of a real algebraic variety can be estimated in terms of the number of monomials in the equations, the known estimates seem to be far weaker than conjectures. For varieties classical from the quantum point of view a strong estimates are obvious. It is very compelling to estimate how much the topology can be complicated by the quantizing deformations.

There seem to be deep relations between the dequantization of algebraic geometry considered above and the results of I. M. Gelfand, M. M. Kapranov and A. V. Zelevinsky on discriminants [1]. In particular, some monomials in a discriminant are related to intersections of hyperplanes in the dequantized polynomial.

Complex algebraic geometry also deserves a dequantization. Especially relevant may be amoebas introduced in [1].

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